

# Music Era and Release-Market Classification

DeepMIR 2026 Fall — Homework 1

---

Chi-Jen Peng (R14725022)

Code, checkpoints, README: <https://github.com/chijenpeng/26-fall-DeepMIR-HW1>

October 5, 2026

## Experimental Setting

---

## Pipeline

- 30 s clip, 24 kHz mono
- frozen encoder → mean + std pooling over time
- StandardScaler → L2 logistic regression
- train = fit, validation = model selection, test = prediction file only

## Hardware / software

---

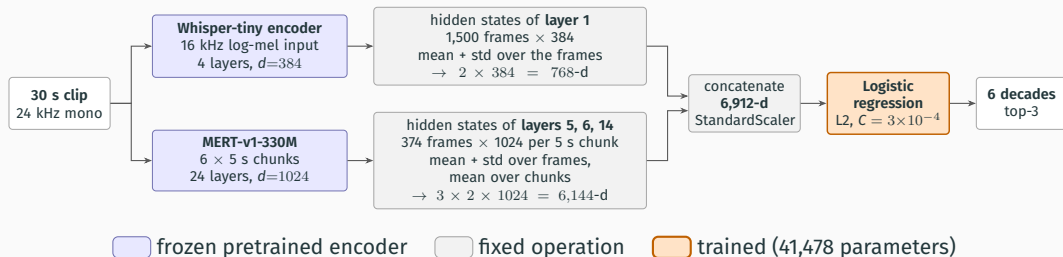
|           |                                                   |
|-----------|---------------------------------------------------|
| GPU       | 2 × NVIDIA RTX A4500 (20 GB)                      |
| CPU / RAM | Intel i9-14900K, 125 GB                           |
| OS        | Ubuntu 26.04                                      |
| Software  | PyTorch 2.14, Transformers 4.45, scikit-learn 1.9 |

---

## Results

---

# Task 1: model



- **Whisper-tiny, layer 1:** shallow layers work best for decade; accuracy drops with depth
- **MERT-v1-330M, layers 5, 6, 14:** best three of a 25-layer sweep; adding Whisper-tiny gives +2.3 points
- **mean + std pooling:** one  $d$ -dim vector per frame  $\rightarrow$  mean and std **over the frames**  $\rightarrow 2 \times d$  values per layer
- Whisper: **1,500 frames** per 30 s clip. MERT: **374 frames** per 5 s chunk (its training length); statistics are taken per chunk, then averaged over the six
- Only the classifier is trained, with strong L2 (1,026 clips, 6,912 features)

# Task 1: release decade

**Model:** frozen encoders + logistic regression

- Whisper-tiny encoder, **layer 1**
- MERT-v1-330M, **layers 5, 6, 14**
- mean + std pooling over time (6,912-d)

Model and layers **chosen by ablation:** [▶ Section 3](#)

| Validation, 132 clips | Top-1        | Top-3        |
|-----------------------|--------------|--------------|
| Best model            | <b>52.3%</b> | <b>86.4%</b> |

## Errors

- **57%** of errors land on a **neighboring decade**
- Hardest: 2000s (recall .23), absorbed by 2010s and 1990s
- Easiest: 1960s and 2010s (one neighbor each)

| True | 1960s | 17        | 2     | 0     | 3     | 0     | 0     |
|------|-------|-----------|-------|-------|-------|-------|-------|
|      | 1970s | 5         | 9     | 3     | 2     | 2     | 1     |
|      | 1980s | 2         | 2     | 13    | 2     | 3     | 0     |
|      | 1990s | 1         | 3     | 2     | 9     | 4     | 3     |
|      | 2000s | 2         | 2     | 1     | 5     | 5     | 7     |
|      | 2010s | 0         | 2     | 0     | 0     | 4     | 16    |
|      |       | 1960s     | 1970s | 1980s | 1990s | 2000s | 2010s |
|      |       | Predicted |       |       |       |       |       |

Counts. Rows = true decade (22 clips each).

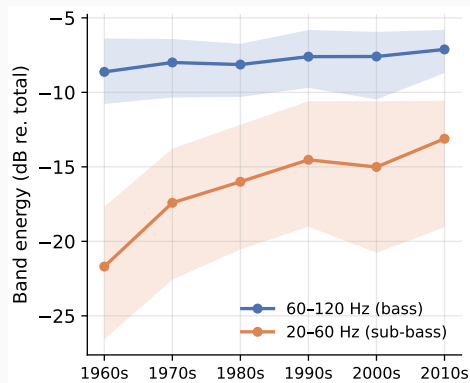
# Why neighbors get confused (1): the low end

**Hypothesis.** Old masters were cut for vinyl, where deep bass makes the stylus jump, so they should carry less sub-bass.

**Analysis.** Long-term average spectrum (Welch) of the 1,026 training clips, split into two bands.

| dB re. total | 60s   | 70s   | 80s   | 90s   | 00s   | 10s   |
|--------------|-------|-------|-------|-------|-------|-------|
| 20-60 Hz     | -21.7 | -17.4 | -16.0 | -14.5 | -15.0 | -13.1 |
| 60-120 Hz    | -8.6  | -8.0  | -8.1  | -7.6  | -7.6  | -7.1  |

- Sub-bass rises by 8.6 dB from the 1960s to the 2010s; the 60-120 Hz band barely moves
- From the 1990s on the low end is nearly identical, so it cannot separate 1990s / 2000s / 2010s



Median and inter-quartile range, training split.

## Why neighbors get confused (2): the loudness war

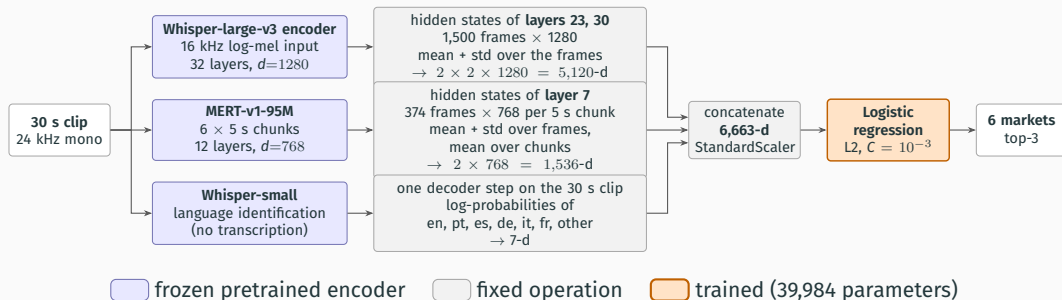
**Hypothesis.** From the 2000s, brick-wall limiters make masters louder and flatten peaks.

| Median per decade, training clips        | 60s   | 70s   | 80s   | 90s   | 00s   | 10s   |
|------------------------------------------|-------|-------|-------|-------|-------|-------|
| RMS level (dBFS)                         | -14.0 | -14.6 | -14.5 | -13.0 | -11.0 | -10.6 |
| Crest factor, peak – RMS (dB)            | 13.9  | 14.5  | 14.5  | 13.0  | 11.0  | 10.6  |
| Dynamic range, loudest 20% blocks (dB)   | 10.6  | 11.2  | 11.3  | 10.7  | 8.9   | 8.7   |
| Longest run pinned at the peak (samples) | 5     | 5     | 6     | 9     | 22    | 26    |
| Samples within 0.5 dB of the peak (%)    | 0.02  | 0.01  | 0.02  | 0.06  | 0.42  | 0.50  |

- 2000s and 2010s clips are 3 dB louder, have 3 dB less crest, and hold peaks at the ceiling. The two are **almost indistinguishable**, matching the 2000s ↔ 2010s confusion
- Production cues only split the data into “old” and “new”; finer distinctions must come from musical content

**Caveat.** The clips are 24 kHz mono, decoded from lossy Opus (YouTube). Resampling, downmixing and lossy coding reshape waveform peaks, so these numbers are trends (ANOVA  $p < 10^{-30}$ ), not exact properties of the original masters. The two sample-level rows are the most affected.

## Task 2: model



- **Whisper-large-v3, layers 23, 30**: deepest layers work best, the opposite of Task 1; this fits language being the main cue
- **MERT-v1-95M, layer 7**: +2 points on top of Whisper (57.8%  $\rightarrow$  59.8%); a music encoder adds what a speech encoder misses
- **Whisper-small language ID**: an explicit 7-d posterior; a small gain (top-3 89.2%  $\rightarrow$  90.2%) because deep Whisper layers already encode language
- Pooling as in Task 1: 1,500 Whisper frames per clip; 374 MERT frames per 5 s chunk

## Task 2: release market

**Model:** frozen encoders + logistic regression

- Whisper-large-v3 encoder, **layers 23, 30**
- MERT-v1-95M, **layer 7**
- Whisper-small language-ID posteriors

Model and layers **chosen by ablation:** [▶ Section 3](#)

| Validation, 102 clips | Top-1        | Top-3        |
|-----------------------|--------------|--------------|
| Best model            | <b>60.8%</b> | <b>90.2%</b> |

### Errors

- **US ↔ UK** is the largest confusion (9 clips)
- **UK is over-predicted:** 23 called UK, only 9 correct
- Brazil (.88) and Spain (.82) are rarely missed
- Germany, Italy: 12 of their 21 errors go to US / UK

| True | US      | 11        | 4  | 0      | 0     | 1       | 1     |
|------|---------|-----------|----|--------|-------|---------|-------|
|      | UK      | 5         | 9  | 0      | 0     | 2       | 1     |
|      | Brazil  | 0         | 0  | 15     | 0     | 0       | 2     |
|      | Spain   | 0         | 2  | 0      | 14    | 1       | 0     |
|      | Germany | 3         | 4  | 1      | 2     | 6       | 1     |
|      | Italy   | 1         | 4  | 0      | 2     | 3       | 7     |
|      |         | US        | UK | Brazil | Spain | Germany | Italy |
|      |         | Predicted |    |        |       |         |       |

Counts. Rows = true market (17 clips each).

# Why English-language markets get confused: language

The dataset has **no language label**. We estimate it with Whisper-small **language identification** on the 798 training clips, on the mixture and on Demucs-separated vocal stems.

| Market  | Home language | Home mixture | Detected as English |        |
|---------|---------------|--------------|---------------------|--------|
|         |               |              | mixture             | vocals |
| US      | English       | 92%          | –                   | –      |
| UK      | English       | 100%         | –                   | –      |
| Brazil  | Portuguese    | 87%          | 11%                 | 12%    |
| Spain   | Spanish       | 57%          | 39%                 | 41%    |
| Germany | German        | 17%          | 83%                 | 82%    |
| Italy   | Italian       | 41%          | 55%                 | 56%    |

Share of training clips whose most likely language (among en, pt, es, de, it) is the home language or English.

- Most German releases and over half of the Italian ones are **sung in English**, so their vocals carry no market cue
- Vocal stems give the same answer: the accompaniment is not what misleads the detector
- Language separates Brazil and Spain. The English-language group has to be split by genre and production style, a much weaker cue
- A release market is where the record was issued, not the artist's nationality or language

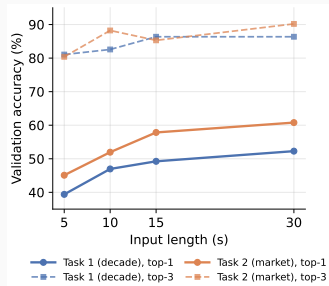
## Optional Experiments

---

## Input length: 5 / 10 / 15 / 30 s

Best models on the first  $k$  seconds only; features re-extracted and classifier re-trained per length.

| Length | Task 1      |             | Task 2      |             |
|--------|-------------|-------------|-------------|-------------|
|        | Top-1       | Top-3       | Top-1       | Top-3       |
| 5 s    | 39.4        | 81.1        | 45.1        | 80.4        |
| 10 s   | 47.0        | 82.6        | 52.0        | 88.2        |
| 15 s   | 49.2        | 86.4        | 57.8        | 85.3        |
| 30 s   | <b>52.3</b> | <b>86.4</b> | <b>60.8</b> | <b>90.2</b> |



- Top-1 rises monotonically on both tasks and has **not saturated** at 30 s; Task 1 top-3 is flat after 15 s
- Task 2 gains more (+15.7 vs. +12.9 points)

## Multiple excerpts per recording

Each 30 s clip is cut into six 5 s chunks. **Chunk-level:** every chunk is its own training sample ( $6\times$  more data) and chunk probabilities are averaged at test time. **Clip-level:** chunk features are averaged first (our default).

| Task | Features                      | Clip-level |       | Chunk-level |       |
|------|-------------------------------|------------|-------|-------------|-------|
|      |                               | Top-1      | Top-3 | Top-1       | Top-3 |
| 1    | MERT-v1-95M, layers 6, 8, 10  | 48.5       | 81.8  | 47.7        | 84.8  |
| 1    | MERT-v1-330M, layers 5, 6, 14 | 50.0       | 87.9  | 50.0        | 86.4  |
| 2    | MERT-v1-95M, layer 7          | 42.2       | 72.5  | 45.1        | 76.5  |
| 2    | MERT-v1-330M, layers 8, 9, 12 | 42.2       | 69.6  | 44.1        | 76.5  |

- Task 1: no gain. Six chunks of one recording are highly correlated, so they add little new information for a linear model
- Task 2: a small gain (+2 to +3 points top-1, +4 to +7 top-3), about 2 to 3 validation clips
- We keep clip-level features for the final models: same accuracy, six times fewer training rows

# Task 1 as year regression

Decades are ordered, so we test a formulation that uses the order.

**Features** (same as the final Task 1 model):

Whisper-tiny encoder layer 1 + MERT-v1-330M layers 5, 6, 14, mean + std pooling, 6,912-d.

**Classification:** L2 logistic regression,  $C = 3 \times 10^{-4}$ .

**Regression:** ridge regression,  $\alpha = 10^4$ , on a decade index (0 = 1960s, 1 = 1970s, ..., 5 = 2010s), rounded to the nearest decade; top-3 = the three nearest decades.

| Formulation                       | Top-1 | Top-3 |
|-----------------------------------|-------|-------|
| Logistic regression (final model) | 52.3  | 86.4  |
| Ridge regression                  | 31.8  | 77.3  |

Mean absolute error of the regression: 0.98 decades.

|      |       |           |       |       |       |       |       |
|------|-------|-----------|-------|-------|-------|-------|-------|
| True | 1960s | 3         | 10    | 6     | 3     | 0     | 0     |
|      | 1970s | 1         | 8     | 8     | 4     | 1     | 0     |
|      | 1980s | 0         | 2     | 10    | 9     | 1     | 0     |
|      | 1990s | 0         | 0     | 5     | 9     | 7     | 1     |
|      | 2000s | 0         | 1     | 5     | 6     | 9     | 1     |
|      | 2010s | 0         | 1     | 1     | 7     | 10    | 3     |
|      |       | 1960s     | 1970s | 1980s | 1990s | 2000s | 2010s |
|      |       | Predicted |       |       |       |       |       |

Ridge regression. Counts; rows = true decade (22 clips each).

## Task 1 as year regression: why it fails

Both rows use the final Task 1 features (Whisper-tiny layer 1 + MERT-v1-330M layers 5, 6, 14); only the predictor differs.

Number of validation clips each formulation assigns to each decade:

| Clips predicted as                | 1960s | 1970s | 1980s | 1990s | 2000s | 2010s |
|-----------------------------------|-------|-------|-------|-------|-------|-------|
| True count                        | 22    | 22    | 22    | 22    | 22    | 22    |
| Logistic regression (final model) | 27    | 20    | 19    | 21    | 18    | 27    |
| Ridge regression                  | 4     | 22    | 35    | 38    | 28    | 5     |

- **Regression is much worse** (−20.5 points top-1): predictions collapse toward the middle, only 4 clips are called 1960s and 5 are called 2010s
- Tuning does not fix it: with a ridge penalty from  $10^2$  to  $10^6$  the best top-1 is 32%; stronger regularization worsens the collapse
- The classifier already learns the order by itself (see the t-SNE slide), so nothing is gained from imposing it

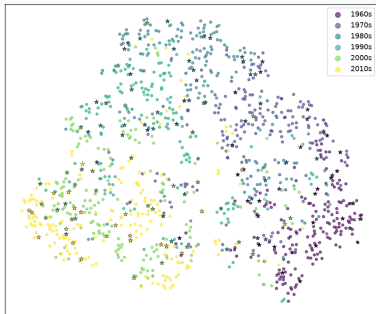
## Task 2: mixture vs. vocals vs. accompaniment

Demucs (htdemucs) separates each clip into vocals and accompaniment. MERT-v1-95M layer 7 features are extracted from each input.

| Input to MERT                    | Top-1       | Top-3       |
|----------------------------------|-------------|-------------|
| Mixture                          | 42.2        | 72.5        |
| Vocals                           | <b>43.1</b> | <b>78.4</b> |
| Accompaniment                    | 33.3        | 61.8        |
| Mixture + vocals + accompaniment | 46.1        | 76.5        |
| Mixture + language ID            | 55.9        | 77.5        |
| Vocals + language ID             | 55.9        | 79.4        |
| Accompaniment + language ID      | 47.1        | 74.5        |

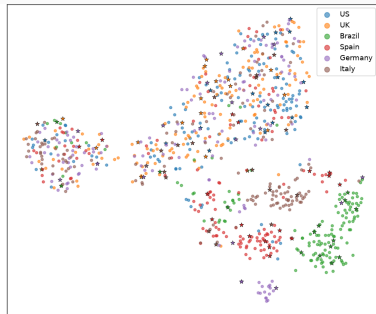
- **Market information is in the vocals:**  
removing them costs 9 points, keeping only them costs nothing
- Vocals halve the US  $\leftrightarrow$  UK confusions (9  $\rightarrow$  4 clips). The accompaniment does not help (9 clips), and its US recall is 18%
- With language ID added, vocals and mixture tie at 55.9%, so the final model uses the mixture: no separation needed at inference

# t-SNE of the best models



**Task 1:** 6-d logit space of the classifier

- Task 1: decades line up along **one arc from the 1960s to the 2010s**, although the classifier was never told they are ordered; 2000s and 2010s overlap
- Task 2: Brazil, Spain and Italy form clusters **before any training**; US, UK and most of Germany share one English-language cloud



**Task 2:** raw feature space, no classifier

Dots = train, stars = validation. Train points in the logit space are fitted; only the stars show generalization.

# Confusion patterns and dataset bias

The detailed analyses are in the Results section. Summary, plus two additional findings:

|                        | Task 1: decade                                                                   | Task 2: market                                                   |
|------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------|
| Confusion pattern      | 57% of errors on a neighboring decade; 2000s ↔ 2010s is the worst pair           | 75% of errors among US, UK, Germany, Italy; UK is over-predicted |
| Bias in the data       | production changes with the era: sub-bass +8.6 dB, crest factor −3 dB after 2000 | English-language releases: 83% of Germany, 55% of Italy          |
| What the model keys on | mastering loudness and bandwidth as well as musical content                      | the language of the vocals rather than the market itself         |

- **Hand-crafted production features** (24-d: band energies, crest factor, limiter statistics) reach 40.9% on Task 1 but only **21.6% on Task 2** (chance 16.7%): within one decade, mastering does not depend on the market
- **High-frequency energy** (8–12 kHz) is 4 to 5 dB lower in the 1960s and 1970s than later; the clips are band-limited to 12 kHz, so anything above that is unavailable

# Ablation

---

## Frozen features vs. fine-tuning

# Audio Language Model

---



## Disclosure

---



## References

---

